INCH-POUND
MIL-M-38510/52D
AMENDMENT 3
28 April 1999
SUPERSEDING
AMENDMENT 2
30 September 1988

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, NOR GATES
MONOLITHIC SILICON, POSITIVE LOGIC

Inactive for new design after 9 August 1996

This amendment forms a part of MIL-M-38510/52D, dated 30 April 1984,
and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.3, add new case outline as follows:

"T F-2A (14-lead, .390" x .260" x .115"), flat package".

PAGE 3

4.2a: Delete "3.1.9 through 3.1.13", and substitute "initial (pre burn-in) electrical parameters (3.1.10) through interim (post burn-in) electrical parameters (3.1.14)".

PAGE 4

4.4.1, insert d and e as follows:

"d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

e. When device types 01 through 03 are qualified by extension (see 4.3.1), these device types will be inspected (QCI) according to the requirements for device types 51 through 53, respectively."

PAGE 6

TABLE I, t_{PHL} and t_{PLH}, delete and substitute as follows:

"

Test	Symbol	Conditions	Device	Limits		Unit
				Min	Max	
Propagation delay times High to low level	t_{PHL}	V_{DD} = 5 V dc C_L = 50 pF See figure 5	All	10	315	ns
			01, 51	10	415	
Low to high level	t_{PLH}		02, 52	13	315	
			03, 04, 53, 54	10	315	

"

TABLE I, t_{THL} and t_{TLH}, Min limits column: Delete "15", "23", and "21" and substitute "10" (3 places).

FSC 5962

AMSC N/A
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TABLE II, Line no. 9, Class B devices, Table III subgroups column: Insert “10 and 11”; Line no. 12: Delete in its entirety.

PAGE 8

FIGURE 1: Delete “Cases A, C and D” and substitute “Cases A, C, D, T, X, and Y” (4 places).

PAGE 15

*TABLE III, I_{IL}, Max limits column: Change “–7.0” to “-700.0”, “-1.0” to “-100.0”, and “-45” to “-100.0”, respectively.
*TABLE III, I_{IH}, Max limits column: Change “7.0” to “700.0”, “1.0” to “100.0”, and “45” to “100.0”, respectively.

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TABLE III, t_{PHL}, t_{PLH}, t_{THL}, t_{TLH}, Subgroup 9, Min limits column: Delete “11”, “14”, “15”, and “23” and substitute “10” (4 places). Subgroup 10 Min limits column: Delete “16”, “21”, “23”, and “34” and substitute “14” (4 places). Subgroup 11, Min limits column: Delete “11”, “14”, “15”, and “23” and substitute “10” (4 places).

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*TABLE III, I_{IL}, Max limits column: Change “–8.0” to “-800.0”, “-1.0” to “-100.0”, and “-45” to “-100.0”, respectively.
*TABLE III, I_{IH}, Max limits column: Change “8.0” to “800.0”, “1.0” to “100.0”, and “45” to “100.0”, respectively.

PAGES 20 AND 21

TABLE III, I_{IH}, Max limits column: change “1.0” to “100.0”, and “45” to “100.0”, respectively.
TABLE III, t_{PHL} and t_{PLH}, Subgroup 9, Min limits column: Delete “30” and substitute “13”. Subgroup 10, Min limits Column: Delete “39” and substitute “18” . Subgroup 11, Min limits column: Delete “15” and “22” and substitute “13” (2 places).
TABLE III, t_{THL} and t_{TLH}, Subgroup 9, Min limits column: Delete “40” and “50” and substitute “10” (2 places). Subgroup 10, Min limits column: Delete “52” and “65” and substitute “14” (2 places). Subgroup 11, Min limits column: Delete “30” and “37” and substitute “10” (2 places).

PAGE 22 THROUGH 24

TABLE III, device types 03 and 04: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y” (3 places)

PAGE 25

TABLE III, device types 03 and 04: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y”
TABLE III, t_{PHL}, t_{PLH}, t_{THL}, t_{TLH}, Subgroup 9, Min limits column: Delete “11”, “15”, and “21” and substitute “10” (3 places). Subgroup 10 Min limits column: Delete “16”, “23”, and “31” and substitute “14” (3 places). Subgroup 11, Min limits column: Delete “11”, “15”, and “21” and substitute “10” (3 places).

PAGE 26 AND 27

TABLE III, device types 03 and 04: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y” (2 places).

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TABLE III, device types 03 and 04: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y”.
*TABLE III, I_{IL}, Max limits column: change “–9.0” to “-900.0”, “-1.0” to “-100.0”, and “-45” to “-100.0”, respectively.
*TABLE III, I_{IH}, Max limits column: change “9.0” to “900.0”, “1.0” to “100.0”, and “45” to “100.0”, respectively.

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TABLE III, device types 03 and 04: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y”.
TABLE III, t_{PHL} and t_{PLH}, Subgroup 9, Min limits column: Delete “11” and substitute “10”. Subgroup 10, Min limits Column: Delete “16” and substitute “14”. Subgroup 11, Min limits column: Delete “11” and substitute “10”.

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TABLE III, device types 03 and 04: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y”.
TABLE III heading, Subgroup 1, Subgroup 2, and Subgroup 3: Delete “1”, “2”, and “3” and substitute “9”, “10”, and “11”, respectively.
TABLE III, t_{THL} and t_{TLH}, Subgroup 9, Min limits column: Delete “15” and “21” and substitute “10” (2 places). Subgroup 10, Min limits column: Delete “23” and “31” and substitute “14” (2 places). Subgroup 11, Min limits column: Delete “15” and “21” and substitute “10” (2 places).

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TABLE III, t_{PHL}, Subgroup 9, Min limits column: Delete “11” and substitute “10”. Subgroup 10, Min limits Column: Delete “16” and substitute “14”. Subgroup 11, Min limits column: Delete “11” and substitute “10”.

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TABLE III heading, Subgroup 1, Subgroup 2, and Subgroup 3: Delete “1”, “2”, and “3” and substitute “9”, “10”, and “11”, respectively.
TABLE III, t_{PLH}, t_{THL}, t_{TLH}, Subgroup 9, Min limits column: Delete “14”, “15”, and “23” and substitute “10” (3 places). Subgroup 10, Min limits column: Delete “21”, “23”, and “34” and substitute “14” (3 places). Subgroup 11, Min limits column: Delete “14”, “15”, and “23” and substitute “10” (3 places).

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*TABLE III, I_{IL}, Max limits column: change “–8.0” to “-800.0”, “-1.0” to “-100.0”, and “-45” to “-100.0”, respectively.
*TABLE III, I_{IH}, Max limits column: change “8.0” to “800.0”, “+1.0” to “100.0”, and “45” to “100.0”, respectively.

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TABLE III, t_{PHL} and t_{PLH}, Subgroup 9, Min limits column: Delete “11” (2 places) and substitute “13” (2 places). Subgroup 10, Min limits column: Delete “16” (2 places) and substitute “18” (2 places). Subgroup 11, Min limits column: Delete “11” (2 places) and substitute “13” (2 places).
TABLE III, t_{THL} and t_{TLH}, Subgroup 9, Min limits column: Delete “15” and “21” and substitute “10” (2 places). Subgroup 10, Min limits column: Delete “23” and “31” and substitute “14” (2 places). Subgroup 11, Min limits column: Delete “15” and “21” and substitute “10” (2 places).

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TABLE III, device types 53 and 54: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y” (2 places).

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TABLE III, device types 53 and 54: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y”.
TABLE III, Test No. 78, pin 13: Delete “5 V”.
TABLE III, Test No. 79, pin 1: Delete “5 V”.
TABLE III, Test No. 80, pin 13: Delete “5 V”.
TABLE III, Test No. 81, pin 1: Delete “5 V”.

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TABLE III, device types 53 and 54: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y” (2 places).
TABLE III, t_{PHL}, t_{PLH}, t_{THL}, t_{TLH}, Subgroup 9, Min limits column: Delete “11” (2 places), “15” and “21” and substitute “10” (4 places). Subgroup 10, Min limits column: Delete “16” (2 places), “23”, and “31” and substitute “14” (4 places). Subgroup 11, Min limits column: Delete “11” (2 places), “15”, and “21” and substitute “10” (4 places).

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TABLE III, device types 53 and 54: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y”.
TABLE III. Test limits, Subgroup 1 and Subgroup 2 columns, Test Nos. 23 through 25: Delete “-25” and “-750”, respectively.

PAGES 47 AND 48

TABLE III, device types 53 and 54: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y” (2 places).

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TABLE III, device types 53 and 54: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y”.
TABLE III, I_{OH1}, Test Nos. 92, 93, and 94: Delete input conditions of “10 V” and substitute “15 V” (18 places).
*TABLE III, I_{IL}, Max limits column: change “–9.0” to “-900.0”, “-1.0” to “-100.0”, and “-45” to “-100.0”, respectively.
*TABLE III, I_{IH}, Max limits column: change “9.0” to “900.0”, “1.0” to “100.0”, and “45” to “100.0”, respectively.

PAGES 50 AND 51

TABLE III, device types 53 and 54: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y” (2 places).
TABLE III, t_{PHL}, t_{PLH}, t_{THL}, t_{TLH}, Subgroup 9, Min limits column: Delete “11” (2 places), “15” and “21” and substitute “10” (4 places). Subgroup 10, Min limits column: Delete “16” (2 places), “23”, and “31” and substitute “14” (4 places). Subgroup 11, Min limits column: Delete “11” (2 places), “15”, and “21” and substitute “10” (4 places).

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4.4.3: Delete 4.4.3c and substitute as follows:

“c. When device types 01 through 03 are qualified by extension (see 4.3.1), these device types will be inspected (QCI) according to the requirements for device types 51 through 53, respectively.”

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA – CC

Preparing activity:
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH
Army - AR, MI, SM
Air Force - 19, 85, 99

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